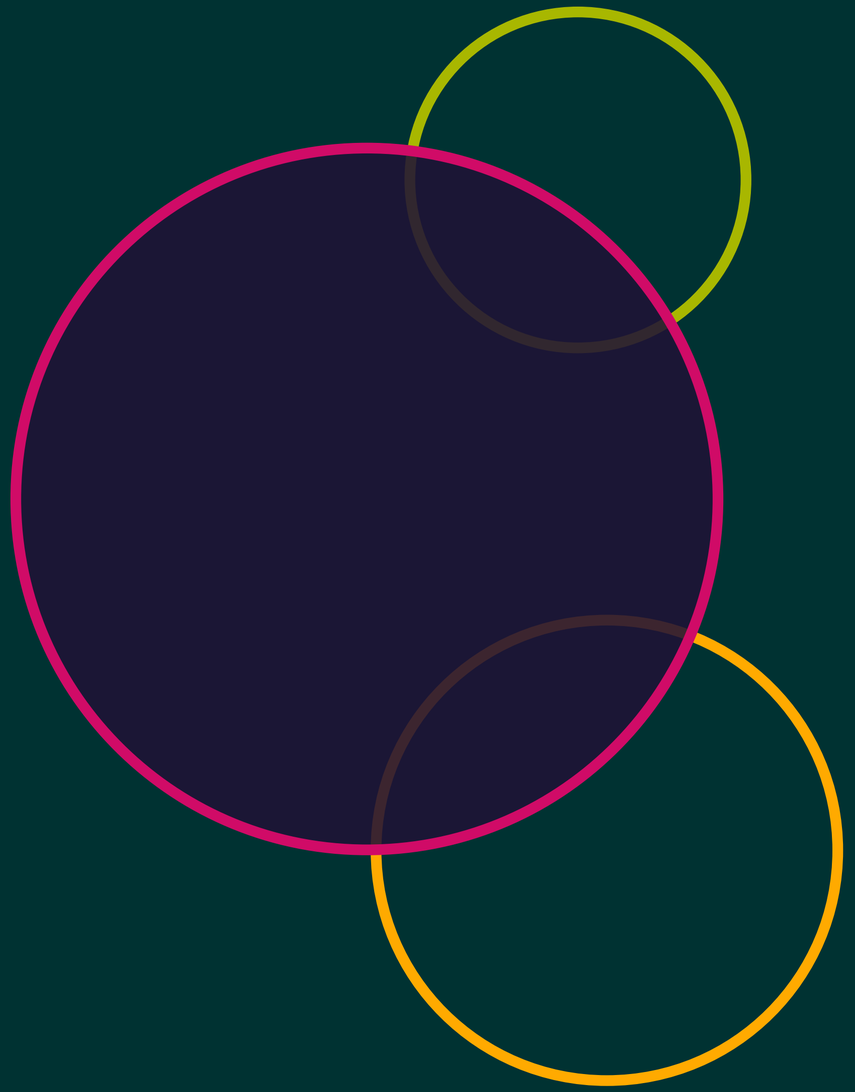
WiredIn

Mojaloop in Production: What You Get, What You Build

Lessons from taking eKash (Rwanda) live and implementing SSIIPS (South Sudan): Where the open-source foundation excels, and where national scale demands custom engineering

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WHO WE ARE

A Kigali-based engineering company building payments and open finance infrastructure.

42
engineers, Kigali, Rwanda

2
national switch programmes

1
1 Mojaloop, 2 deployments

eKash - Rwanda	Mojaloop-based national switch (RNDPS 2.0), built for RSwitch Ltd. Delivered and operated in production.
SSIIPS - South Sudan	Mojaloop-based national instant payment system for the Bank of South Sudan. In implementation.
In the community	Open Banking Workstream Lead; certified channel partner for Tazama, the open-source fraud and risk management system.

WHAT WORKS WELL OUT OF THE BOX

For early-stage development and sandbox testing, the open-source foundation removes most of the initial friction.

✓

Infrastructure deployment base

For standard environments, the Kubernetes and Helm chart configurations give you a streamlined Infrastructure-as-Code path to standing up the core microservices.

Significantly reduces the friction of getting the core switch running.

✓

The Testing Toolkit (TTK)

Invaluable for validating basic API compliance, and reliable for simulating DFSP behavior throughout initial testing and sandbox phases.

Our Switch Deployment comfort tests in both programmes started from the TTK.

GAP 1: CUSTOM DEVELOPMENT REQUIRED

Mojaloop clears and settles. A national scheme needs a full business capability set. Below is what was custom developed in our deployments.

1

ISO 20022 connector with XMLDSig

Country-specific for RNDPS: message transformation plus XML digital signatures for integrity and non-repudiation.

2

Hub Admin and DFSP portals

Operational front-ends for the hub operator and self-service portals for participants - onboarding, configuration, visibility.

3

Auto-reconciliation, end-to-end

Automated matching across the full transaction path, so positions are proven continuously rather than investigated after the fact.

4

Dispute management

Structured workflows for exceptions, refunds and scheme-rule disputes between participants - a scheme obligation, not an optional extra.

5

RTGS system integration

Settlement finality means connecting the switch to the central bank's RTGS - its formats, its windows, its cut-offs.

6

Tazama FRMS integration

Real-time fraud and risk screening wired into the transaction flow from day one, using the open-source Tazama platform.

GAP 2: CUSTOM DEVELOPMENT REQUIRED

Out-of-the-box tooling tells you whether pods are healthy. A national switch must know whether payments are healthy.

What you get

- Basic infrastructure health metrics
- Pod, node and service-level status
- Generic dashboards and alerts

What we had to build

- Business-centric dashboards tracking financial SLAs
- Per-DFSP transaction success, latency and failure views
- Settlement and reconciliation health at a glance
- Alerting tied to scheme obligations, not CPU

GAP 3: CUSTOM DEVELOPMENT REQUIRED

Default Helm charts do not give you a hardened, multi-region production environment.

1

Multi-region topology was custom engineered

True high availability and disaster recovery means designing active-active or active-passive clusters across regions.

2

Stateful services are the hard part

Kafka and the core databases need carefully designed cross-region replication. Consistency, failover behavior and recovery-point objectives are real design decisions, not defaults.

OUR INSIGHT

Open source eliminates vendor lock-in but replaces the license fee with a capability bill.

What you escape

Vendor lock-in, per-transaction licence economics, and roadmaps set by someone else's commercial priorities.

What you take on

The obligation to independently secure, patch, and scale the system - forever.

That demands a massive, sustained investment in team capacity:

Local DevOps engineers	who own the clusters, pipelines, upgrades and incident response
Security specialists	who track CVEs, run the patch cadence and harden the infrastructure
Software engineers	who extend, integrate and performance-tune the switch itself

THANK YOU

Q & A

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